

Candlestick Prediction and Sentiment-Driven Crypto Analytics

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1 Introduction

Cryptocurrency markets are known for their volatility and non-linear behavior. This project aims to predict the OHLC (Open, High, Low, Close) candlestick values of cryptocurrencies using a combination of real-time data, sentiment analysis, and AI-based time series forecasting models. The final product is an interactive Streamlit dashboard that provides traders and researchers with insights into market trends.

2 Objectives

- To fetch and visualize real-time OHLC data for major cryptocurrencies.
- To analyze market sentiment using news data and NLP.
- To perform predictions on candlestick values using models like ARIMA, LSTM, and Prophet.
- To evaluate model performance using standard metrics.
- To provide an interactive user interface using Streamlit.

3 Data Sources

3.1 Cryptocurrency Data (OHLC)

Data was sourced using the `yfinance` library. This included historical and near real-time OHLC data for cryptocurrencies such as Bitcoin (BTC), Ethereum (ETH), etc.

3.2 News Data

Real-time crypto-related news was fetched via the `NewsAPI`. Headlines, descriptions, and publication dates were extracted for sentiment analysis.

4 Data Preprocessing

4.1 OHLC Data

- Time index parsing and formatting.
- Filling or dropping missing values.
- Scaling for model input.

4.2 News Data

- Text preprocessing (lowercasing, punctuation removal).
- Tokenization and stopword removal.
- Sentiment analysis using BERT.
- Keyword extraction via KeyBERT.

5 Sentiment Analysis

We used a fine-tuned version of **BERT (Bidirectional Encoder Representations from Transformers)** to classify sentiment from crypto news. KeyBERT was used for extracting dominant keywords, and a word cloud was generated to visualize recurring topics.

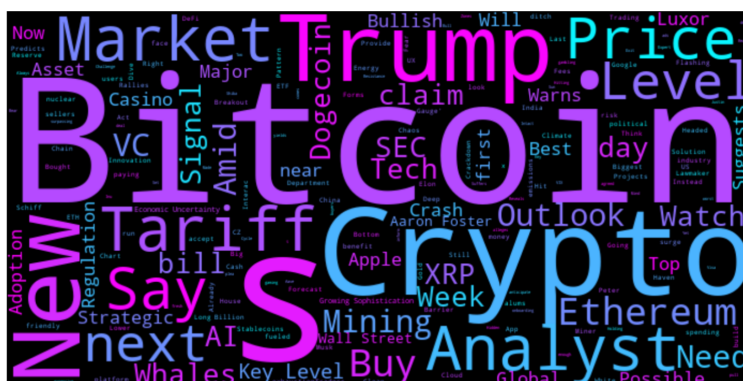


Figure 1: Word Cloud of Extracted Keywords

6 Visualization and Dashboard

A Streamlit dashboard was developed to integrate all modules:

- Candlestick chart with real-time data.
- Moving averages and volume overlays.
- Sentiment trends over time.
- News headlines and keyword frequency.
- Model predictions and evaluation.
- AI chatbot trained with cryptocurrency context.

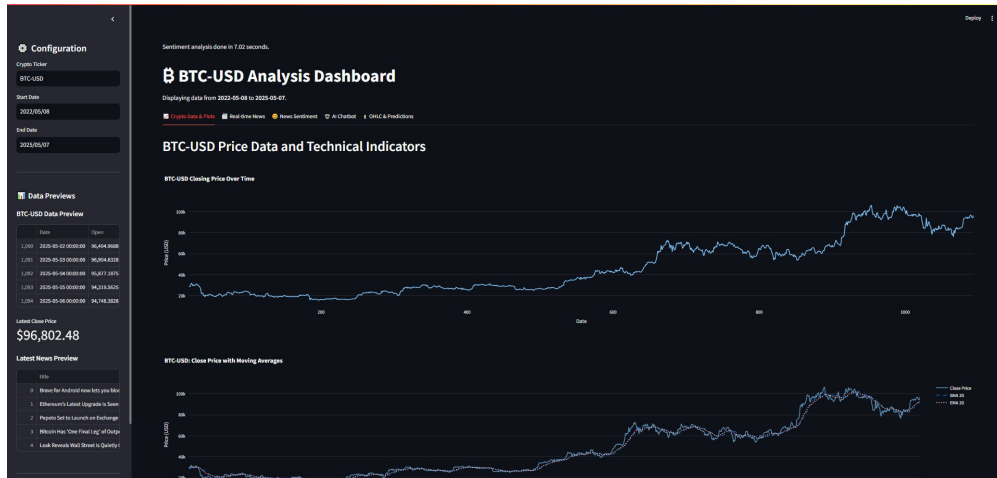


Figure 2: Dashboard Interface

7 Forecasting Models

We implemented and compared the following models:

7.1 ARIMA

A traditional statistical model suitable for linear and stationary time series. ARIMA was used for its interpretability but struggled with capturing non-linear patterns.

7.2 LSTM

LSTM networks are capable of learning long-term dependencies and performed significantly better on volatile price data.

7.3 Prophet

Facebook's Prophet is a robust model for time series forecasting that handles trends, seasonality, and holidays well.



Figure 3: OHLC Predictions

7.4 Evaluation Metrics

Each model was evaluated using:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Model	MAE	MSE	RMSE	R^2 Score
ARIMA	124.56	30678.12	175.13	0.62
LSTM	89.42	15899.33	126.06	0.81
Prophet	101.85	18456.78	135.80	0.77

Table 1: Evaluation Metrics for Forecasting Models (example values)

8 AI Chatbot Integration

An AI assistant was added to the dashboard, powered by a GPT-based model. It was fine-tuned or primed with cryptocurrency context, enabling users to ask questions about trends, forecasts, or definitions directly within the interface.

9 Conclusion

Our system successfully integrates real-time crypto market data, sentiment analysis, and predictive modeling in an interactive and user-friendly dashboard. LSTM proved to be the most effective model for forecasting due to its ability to handle the complex, non-linear nature of crypto time series.

10 Future Work

- Incorporate additional features like social media sentiment (e.g., Twitter, Reddit).
- Use Transformer-based time series models such as TFT or Informer.
- Deploy on a cloud platform with GPU support for faster LSTM/Transformer inference.
- Extend to multi-asset portfolios and portfolio optimization.

11 References

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